

The Decision Boundary 🌐

The decision boundary is a key concept in understanding how a logistic regression model makes its final predictions. It is the line or curve that separates the different predicted classes.

From Probability to Prediction

To convert the model's probability output into a discrete prediction (0 or 1), a **threshold** is used.

- The model's output is the probability that $y = 1$:

$$f_{\vec{w},b}(\vec{x}) = P(y = 1|\vec{x})$$

- A common threshold is **0.5**.

- **Prediction Rule:**

- If $f_{\vec{w},b}(\vec{x}) \geq 0.5$, predict $\hat{y} = 1$.
- If $f_{\vec{w},b}(\vec{x}) < 0.5$, predict $\hat{y} = 0$.

By analyzing the Sigmoid function, $g(z)$, we know that $g(z) \geq 0.5$ only when its input, z , is greater than or equal to 0.

- This simplifies the prediction rule significantly:
 - Predict $\hat{y} = 1$ when $z = \vec{w} \cdot \vec{x} + b \geq 0$.
 - Predict $\hat{y} = 0$ when $z = \vec{w} \cdot \vec{x} + b < 0$.

The **decision boundary** is the line defined by the equation where the model is exactly on the fence:

$$\vec{w} \cdot \vec{x} + b = 0$$

Types of Decision Boundaries

Linear Decision Boundary

- When using simple features (e.g., x_1, x_2), the decision boundary is a straight line.
- **Example:**
 - Model with two features: $z = w_1x_1 + w_2x_2 + b$.
 - If parameters are $w_1 = 1, w_2 = 1, b = -3$.
 - The decision boundary is the line where $x_1 + x_2 - 3 = 0$, or $x_1 + x_2 = 3$.

Non-Linear Decision Boundary

- By using **polynomial features**, logistic regression can learn complex, non-linear decision boundaries.
- **Example:**
 - Model with polynomial features: $z = w_1x_1^2 + w_2x_2^2 + b$.
 - If parameters are $w_1 = 1, w_2 = 1, b = -1$.
 - The decision boundary is the curve where $x_1^2 + x_2^2 - 1 = 0$, or $x_1^2 + x_2^2 = 1$. This is a circle.
- Higher-order polynomial terms can create even more complex shapes, like ellipses or other intricate curves, allowing the model to fit very complex datasets.